

D5: Formal Languages, DFA, and Turing Machines



Sec 16: Fr 12:30pm-1:30pm FXB 1024
IA: Nick Frieler

Agenda

- ▶ Formal languages
 - ▶ Many definitions
- ▶ (Deterministic) finite automata
 - ▶ Definition
- ▶ Turing machines
 - ▶ Definition
 - ▶ Church-Turing thesis

Formal Languages

Time to get try-lingual

- ▶ An **alphabet** is a *nonempty, finite* set of symbols
 - ▶ a, b, c, d, ..., z
 - ▶ +, -, x, /, 0, 1, 2, ..., 9
- ▶ A **string** over an alphabet Σ is a *finite, ordered* sequence of symbols from Σ
 - ▶ If $\Sigma = \{a, \dots, z\}$, then “hello” is a string over Σ
 - ▶ If $\Sigma = \{+, -, x, /, 0, \dots, 9\}$, then “376+281+370” is a string over Σ
 - ▶ Σ^* is the set of *all* strings over Σ
- ▶ A **language** L over an alphabet Σ is a subset of Σ^*
 - ▶ The set of all decimal representations of even numbers is a language over $\{0, \dots, 9\}$

(Deterministic) Finite Automata

Desperately Following Arrows

- ▶ A *deterministic finite automaton (DFA)* is a 5-tuple $M = (\Sigma, Q, q_0, F, \delta)$
 - ▶ Σ is the input alphabet
 - ▶ Q is the finite set of the automaton's states
 - ▶ $q_0 \in Q$ is the initial state
 - ▶ $F \subseteq Q$ is the subset of accepting states
 - ▶ δ is the transition function, it maps from $Q \times \Sigma \rightarrow Q$
 - ▶ Because a DFA is **deterministic**, δ must be a *function* (exactly one output for each input)
- ▶ The *language L of a finite automaton M* is the set of strings it accepts. We also say
 - ▶ M decides L if and only if M accepts every string in L and reject every other string

Turing Machines

Please just one more definition

- ▶ A *Turing machine* is a 7-tuple $M = (\Sigma, \Gamma, Q, q_{\text{start}}, q_{\text{acc}}, q_{\text{rej}}, \delta)$
 - ▶ Σ is the input alphabet
 - ▶ Γ is the tape alphabet (contains Σ and a blank symbol $\perp$)
 - ▶ Q is the finite set of states
 - ▶ q_{start} is the initial state
 - ▶ $q_{\text{acc}}, q_{\text{rej}} \in Q$ are the accept state and reject state, respectively (must be distinct)
 - ▶ We say $F = \{q_{\text{acc}}, q_{\text{rej}}\}$ is the set of *final states*
 - ▶ δ is the transition function

$$\delta: \underbrace{(Q \setminus F)}_{\text{Non-terminal state}} \times \underbrace{\Gamma}_{\text{Cell contents}} \rightarrow \underbrace{Q}_{\text{Next state}} \times \underbrace{\Gamma}_{\text{Symbol to write}} \times \underbrace{\{L, R\}}_{\text{Where to move the head}}$$

Ok but what does a TM do?

- ▶ The input is given on the TM's tape
- ▶ The TM head points to the first cell of the tape, starting in state q_{start}
- ▶ The TM repeatedly applies its transition function δ until it reaches one of q_{acc} or q_{rej}
 - ▶ If this never happens, the TM runs forever
- ▶ Because a TM M might not halt on an input x , we have three cases:
 - ▶ $M(x)$ eventually *accepts*
 - ▶ $M(x)$ eventually *rejects*
 - ▶ $M(x)$ runs forever (or “*loops*”)

Ok but what does a TM do?

- ▶ The *language* $L(M)$ of a Turing machine M is the set of strings M accepts
- ▶ A TM *decides* a language L if
 - ▶ M accepts every string in L
 - ▶ M rejects every string not in L
 - ▶ NO LOOPING
- ▶ A TM *recognizes* a language L if
 - ▶ M accepts every string in L
 - ▶ M rejects or loops on every string not in L

Church-Turing thesis

- ▶ A function can be computed by an “effective” or “mechanical” procedure (i.e., an algorithm) *if and only if* it can be computed by some Turing machine
 - ▶ This is not a statement that we can prove!
 - ▶ This is essentially a definition of what *computation* is



Any questions before
discussion problems?

Discussion Problems

Q1: Drawing Forty Arrows

True or False: there is a DFA that decides the following language:

$L = \{x: x \text{ is a binary string of length 4 that starts with two 1s}\}$

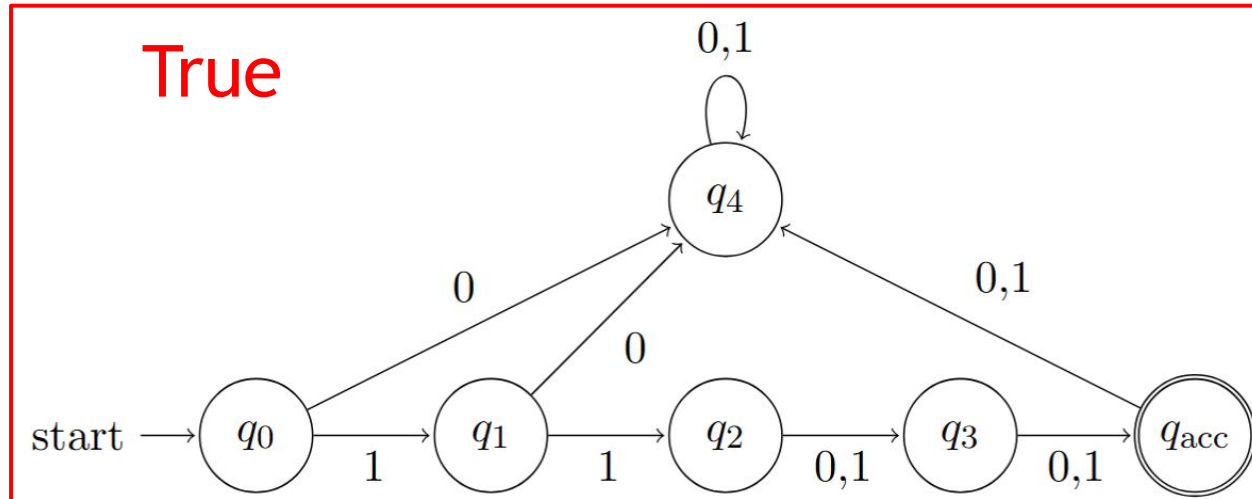
Justify your answer

Q1: Drawing Forty Arrows

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Justify your answer



Q2: Is that a palindrome?

True or False: there is a DFA that decides the following language:

$L = \{x: x \text{ is palindrome}\}$ where the language Σ has at least 2 elements

Justify your answer

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Justify your answer

- ▶ **False**... but why?
- ▶ Suppose there is a DFA M (with s states) that decides L
- ▶ Let x and y be two (distinct) symbols in Σ
- ▶ Consider the states M is in after processing the strings $x^1, x^2, \dots, x^s, x^{s+1}$
 - ▶ By the pigeonhole principle, M must be in the same state after processing at least two of these strings, say x^i and x^j where $i \neq j$
- ▶ What happens when we run M on the strings $x^i y x^i$ and $x^j y x^j$?
 - ▶ It has to either reject both or accept both
 - ▶ **CONTRADICTION** $\rightarrow$ no DFA decides L

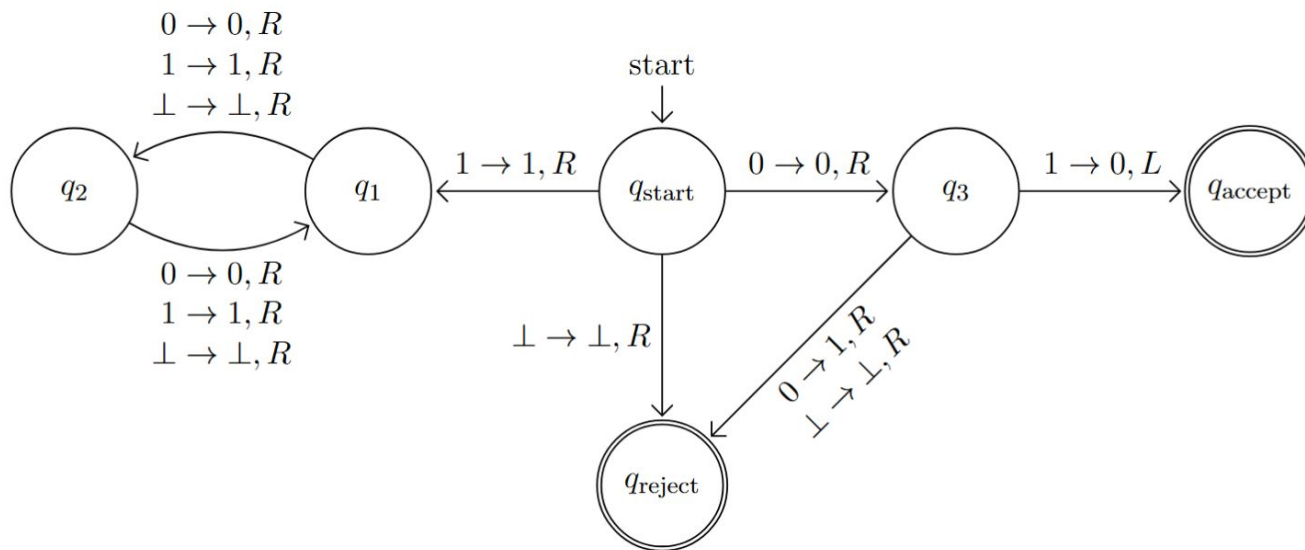
Attendance



<https://tinyurl.com/2za4dcmm>

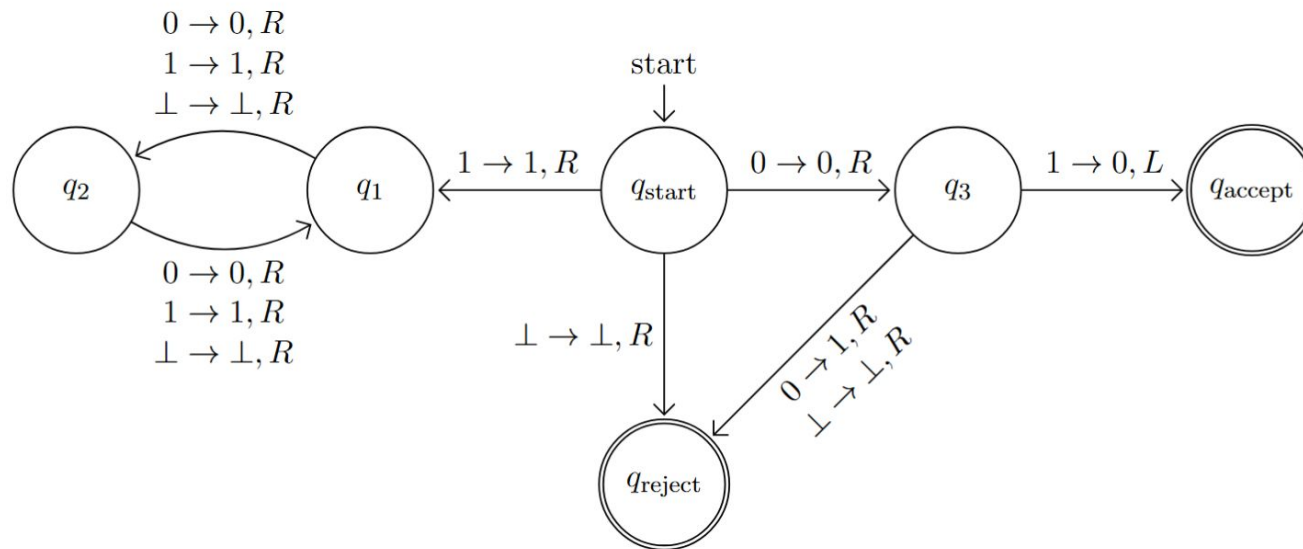
Q3: Decisions decisions

What language does the following Turing machine accept, if any?



Q3: Decisions decisions

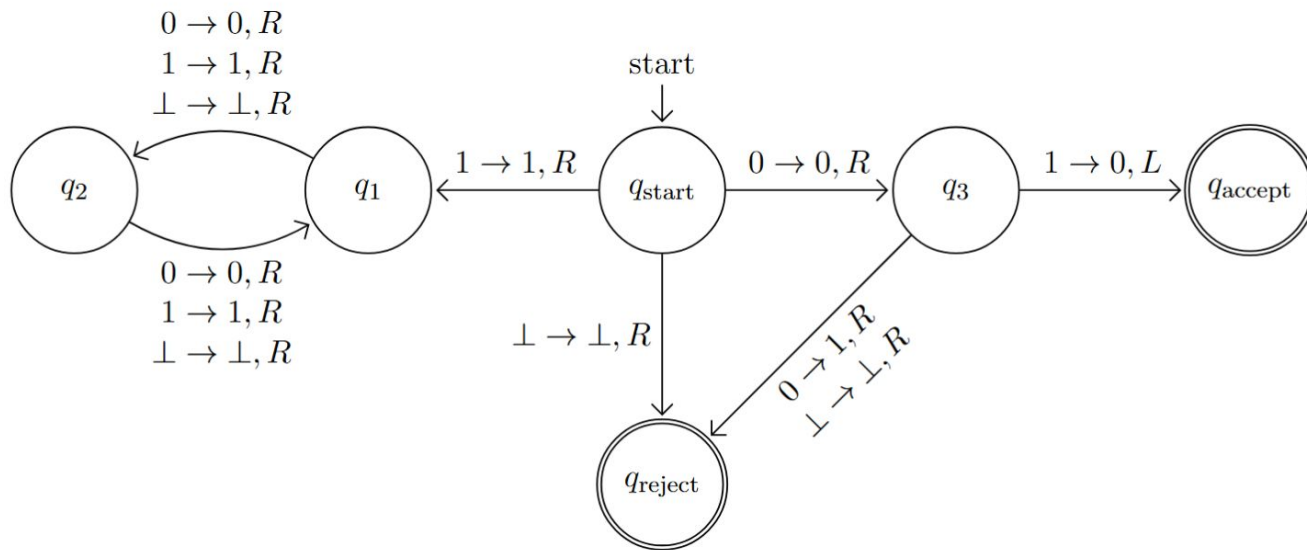
What language does the following Turing machine decide, if any?



This Turing machine **does not decide any language!** For example, it **loops on input “11”**

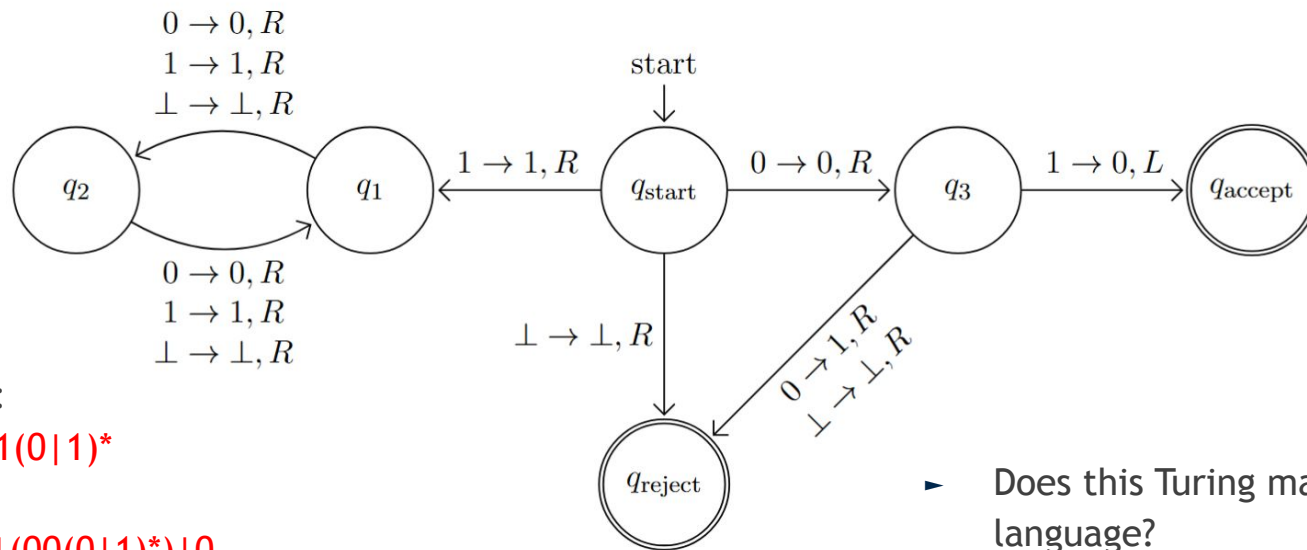
Q4: Pattern recognition

What inputs does this Turing machine (same as last question) accept? reject? loop on?



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What inputs does this Turing machine (same as last question) accept? reject? loop on?



► Accept:

► $01(0|1)^*$

► Reject:

► $\varepsilon | (00(0|1)^*) | 0$

► Loop:

► $1(0|1)^*$

► Does this Turing machine recognize a language?

► **Of course! Every Turing machine recognizes a language**

Q5: Decidedly decidable

Let A and B be two decidable languages. Given access to a Turing machine M_A that decides A and a Turing machine M_B that decides B , what languages based on A and B can you come up with that are also decidable?

Q5: Decidedly decidable

Let A and B be two decidable languages. Given access to a Turing machine M_A that decides A and a Turing machine M_B that decides B , what languages based on A and B can you come up with that are also decidable?

- ▶ $A \cup B$
 - ▶ $x \in A \cup B \Leftrightarrow M_A(x)$ accepts OR $M_B(x)$ accepts
 - ▶ $A \setminus B$
 - ▶ $x \in A \setminus B \Leftrightarrow M_A(x)$ accepts AND $M_B(x)$ rejects
 - ▶ $B \setminus A$
 - ▶ $x \in B \setminus A \Leftrightarrow M_A(x)$ rejects AND $M_B(x)$ accepts
 - ▶ $A \cap B$
 - ▶ $x \in A \cap B \Leftrightarrow M_A(x)$ accepts AND $M_B(x)$ accepts
- ▶ We can also decide the complements of these languages